POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE | Full Marks : 100
Course: Computer Architecture (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate Computer organization and Computer Architecture in detail with examples. 7

OR

What are the functional requirements of computer and how they support in fulfilment of structure of computer? Explain in detail.

b) Define RTL. Write about different shift operations and also express them in RTL. 8

OR

Write VHDL code for full adder with two half adder and a OR gate.

2. a) What is instruction Cycle? Explain different stage of instruction cycle. 7

b) Define control memory. Explain the working of microprogrammed control unit. Differentiate between Horizontal and Vertical microprogrammed control unit. 8

3. a) Multiply 13*8 using booth's algorithm. 8

b) Explain the restoring division algorithm with flowchart. 7

4. a) Explain different data hazards present during instruction pipeling? Suggests their solutions. 8

b) Why mapping is needed in cache? Explain different cache write policy and cache replacement algorithm used. 7

5. a) Briefly explain the various memory storage devices of memory organization. 7

b) Explain the structure of an I/O module and major functions that I/O module performs. 8

6. a) Describe in brief about Flynn's Taxonomy. Also explain how parallelism can be experienced in uniprocessor system. 7

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b) Explain different types of hardware performance issues in multicore computers. 8

7. Write short notes on: **(Any two)** 2×5

a) GPU and TPU

b) Design Principle for Modern System

c) RISC and CISC architecture

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